



Peerachet Khanitson

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SUMMARY

Computer Engineering graduate with a strong focus on Artificial Intelligence and Machine Learning. I have hands-on experience analyzing complex datasets and building predictive models for real-world scenarios. I am highly motivated to transform data into actionable insights, and eager to apply my technical knowledge to develop intelligent solutions while growing as an AI Engineer.

EDUCATION

Mae Fah Luang University - School of Applied Digital Technology. Aug 2022 - Apr 2026
Bachelor of Engineering - Computer Engineering | **GPAX: 3.68**

PROJECT EXPERIENCE

Senior Project (Machine Learning model training) <https://github.com/peerachetkns/PM2.5>

- **Project Overview:** Developed an end-to-end multi-class machine learning pipeline to classify PM2.5 levels into 5 health-based categories according to the Thai Air Quality Standard.
- **Data Engineering:** Processed massive hourly meteorological and pollutant datasets from 12 monitoring stations. Executed data cleaning, missing value imputation (Median & Linear Interpolation), outlier capping, and feature engineering, including trigonometric encoding for cyclic wind direction and Min-Max scaling.
- **Imbalance Handling:** Systematically evaluated multiple oversampling algorithms (SMOTE, Borderline-SMOTE, SMOTE-Tomek, ADASYN, KMeans-SMOTE) using both selective and full equalization strategies to resolve severe class imbalance.
- **Model Evaluation & Results:** Benchmarked 8 distinct ML and DL architectures (including CNN, LSTM, KNN, RF, GB, and SVM). The optimal model, an Extra Trees ensemble with Borderline-SMOTE, achieved an outstanding 92.7% Accuracy, 92.7% F1-Score, and 99.3% ROC-AUC.

WORK EXPERIENCE

Mitsubishi Electric Thai Auto-Parts Co., Ltd. Internship Jan - Apr 2026 (4 Months)

- **System Development:** Digitized workflows using a centralized Power Apps solution.
- **Process Automation:** Automated syncing via Power Automate to reduce manual errors.
- **Data & UI/UX:** Structured data and designed a minimalist UI to boost operational efficiency.

SKILLS

Technical Skills

- **Programming Languages:** Python, C, Java
- **ML & DL:** Classification, Regression, Clustering, Ensemble (RF, ET, GB), Deep Learning (CNN, LSTM)
- **Model Evaluation:** Cross-Validation, Bias/Variance, Key Metrics (Accuracy, Precision, F1, ROC-AUC)
- **Data Analysis & Prep:** EDA, Data Cleaning, Feature Engineering, Transformation, Imbalanced Handling
- **CV & Image Processing:** Image Enhancement, Spatial/Frequency Filtering, Restoration, Color Processing
- **Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow/Keras
- **Tools & Platforms:** VS Code, Jupyter Notebook, Google Colab, Git & GitHub, Microsoft Power Platform (Power Apps, Power Automate)
- **Databases:** MySQL, Firebase Firestore
- **Web and Mobile App Development:** HTML, CSS, JavaScript, Flutter (Dart)

Soft Skills

- **Collaboration** – Worked collaboratively with peers on a Machine Learning project.
- **Adaptability & Active Learning** – Proactive in acquiring new knowledge and skills independently.
- **Problem Solving** – Skilled at debugging and resolving real-world code issues in project work.
- **Communication** – Able to communicate effectively in both spoken and written English.